Compile Flower, Fruit, and Seed count data sheets

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```
#Load Packages
library(tidyverse)
library(janitor)
library(here)
library(lubridate)
library(reshape2)
library(stringr)
```

Load and Clean Data

NOTE: All date data should be standardized prior to importing into R. In Excel, set format to YYYY-MM-DD for consistency.

NOTE 2: All paths are relative using the 'here' package.

Germination Data

- Add 'Date Sown' column and data in as well. - for AC germination, plant number must be FLIPPED (i.e., plants 1,2,3,4,5,6,7,8,9,10 become plants 10,9,8,7,6,5,4,3,2,1, respectively)

```
#Load csv

#Skip first nonsense row (the second row is the header)
AC_germ_df1 <- read_csv(here::here("data_sheets", "germination_checks_2021-2022", "AC_germ_data.csv"), s
  rename(Block = 'Block (1-3)', Transect = 'Transect (1-9)', Sequence = 'Sequence (1-321)', Donor = "Pol
  filter(!Stage == "Flow Date")   #Remove empty flowering date rows


#Reshape data sheet to 'long' format
AC_germ_df2 <- AC_germ_df1 %>%
  pivot_longer(cols = c('1':'10'),
               names_to = 'Plant_ID',
               values_to = 'Germ_Date')


##Fix typos in date values
##Excel auto-populated year as 2022 when in fact the year should be 2021!
unique(AC_germ_df2$Germ_Date)
```

```
##  [1] "2022-12-05"
##  [2] NA
##  [3] "2022-11-03"
##  [4] "Dead 12/5, plug missing"
##  [5] "11/3/2021, disturbed by animal (deer)"
##  [6] "disturbed by animal (deer)"
##  [7] "11/3 (n=2, n=1 on 12/5)"
##  [8] "12/5/2021, disturbed by animal (deer)"
```

```
##  [9] "12/5, ? plug buried"
## [10] "? disturbed by animal (deer)"
## [11] "12/5/2021, ?"
```

```
AC_germ_df3 <- AC_germ_df2 %>%
  mutate(Germ_Date = replace(Germ_Date, Germ_Date %in% c("Dead 12/5, plug missing", "disturbed by animal
mutate(
  Germ_Date = case_when(
    Germ_Date %in% c("2022-12-05", "12/5, ? plug buried", "12/5/2021, ?", "12/5/2021, disturbed by anima
    Germ_Date %in% c("2022-11-03", "11/3/2021, disturbed by animal (deer)", "11/3 (n=2, n=1 on 12/5)") 

#check again
unique(AC_germ_df3$Germ_Date)
```

```
## [1] "2021-12-05" NA           "2021-11-03"
```

```
#Use sum(is.na(AC_germ_df2$Date)) to compare change in number of NA values


#
#Adjust Donor and Recipient names for easy sorting - three digits (leading zeros)
#Specify Germ_Date as 'date' class variable
AC_germ_df4 <- AC_germ_df3 %>%
  mutate(Recipient_num = sprintf("%03d", (parse_number(Recipient)))) %>%
  mutate(Recipient = as.character(str_c("AC_", Recipient_num))) %>%
  mutate(Donor_num = sprintf("%03d", (parse_number(Donor)))) %>%
  mutate(Donor = as.character(str_c("AC_", Donor_num))) %>%
  mutate(Germ_Date = as_date(Germ_Date)) %>% #set variable to class = date
  select(Block, Transect, Sequence, Donor, Donor_num, Recipient, Recipient_num, Plant_ID, everything())
  select(-c(Stage, Donor_num, Recipient_num))


#
##Add in Sow Date (lab sown) and Plant date (in field)
AC_germ_df5 <- AC_germ_df4 %>%
  mutate(Sow_Date = as_date("2021-10-27")) %>%
  mutate(Plant_Date = as_date("2021-11-04"))
#Some plugs may have been sown on November 5, 2022 (information on which transects not recorded)


#
## Flip Plant numbers
## (AC germination data only) - the orientation in which plants were identified and checked for floweri
AC_germ_df6 <- AC_germ_df5 %>%
  mutate(Plant_ID = as.numeric(case_when(
    Plant_ID == "1" ~ "10",
    Plant_ID == "2" ~ "9",
    Plant_ID == "3" ~ "8",
    Plant_ID == "4" ~ "7",
    Plant_ID == "5" ~ "6",
    Plant_ID == "6" ~ "5",
    Plant_ID == "7" ~ "4",
    Plant_ID == "8" ~ "3",
    Plant_ID == "9" ~ "2",
    Plant_ID == "10" ~ "1"))) %>%
```

```
  arrange(Block, Transect, Sequence, Recipient, Plant_ID) #Arrange by Transect, Recipient, Plant_ID
```

Add in watered transect germination data

```
#AC
## Add in watered transects
## *Note*: Only 8 seeds sown in these segments instead of 10
AC_germ_W1 <- read_csv(here::here("data_sheets", "germination_checks_2021-2022", "watered_AC_germ.csv")
  rename(Sequence = 'Seq (1-42)', Donor = "Donor_ID", Recipient = "Recipient_ID", Stage = 'Date (mm/dd)
  filter(!Stage == "Flow") %>%    #Remove empty flowering date rows
  mutate(`2` = as_date(`2`), `5` = as_date(`5`), `8` = as_date(`8`)) #these columns not recognized as d

#
#Reshape data sheet to 'long' format
AC_germ_W2 <- AC_germ_W1 %>%
  pivot_longer(cols = c('1':'8'),
               names_to = 'Plant_ID',
               values_to = 'Germ_Date')

#
##Fix typos in date values
unique(AC_germ_W2$Germ_Date) #no typos here
```

```
## [1] "2022-12-05" "2022-11-03" NA
```

```
AC_germ_W3 <- AC_germ_W2 %>%
  mutate(Germ_Date = case_when(
    Germ_Date == "2022-12-05" ~ "2021-12-05",
    Germ_Date == "2022-11-03" ~ "2021-11-03"))


#
#Adjust Donor and Recipient names for easy sorting - three digits (leading zeros)
#Specify Germ_Date as 'date' class variable
AC_germ_W4 <- AC_germ_W3 %>%
  mutate(Recipient_num = sprintf("%03d", (parse_number(Recipient)))) %>%
  mutate(Recipient = as.character(str_c("AC_", Recipient_num))) %>%
  mutate(Donor_num = sprintf("%03d", (parse_number(Donor)))) %>%
  mutate(Donor = as.character(str_c("AC_", Donor_num))) %>%
  mutate(Germ_Date = as_date(Germ_Date)) %>% #set variable to class = date
  mutate(Plant_ID = as.numeric(Plant_ID)) %>% #set Plant ID to numeric for sorting purposes
  select(-c(Stage, Co., Donor_num, Recipient_num))


#
##Add in Sow Date & Update Sequence number & add block ('W')
AC_germ_W5 <- AC_germ_W4 %>%
  mutate(Sow_Date = as_date("2021-10-27")) %>%
  mutate(Plant_Date = as_date("2021-11-04")) %>%
  mutate(Sequence = (Sequence + 321)) %>%
  mutate(Block = "W") %>%
  select(Block, Transect, Sequence, Donor, Recipient, Plant_ID, everything()) %>%
  arrange(Transect, Sequence, Recipient, Plant_ID) #Arrange by Transect, Recipient, Plant_ID
## Flipping Plant numbers not required for watering transects - plugs planted top-to-bottom for ID 1 th
```

```r
#
## Combine watered and main experiment germination data
AC_germ_full <- rbind(AC_germ_df6, AC_germ_W5)

#write_csv(AC_germ_full, here::here("data_sheets", "compiled_sheets", "AC_germination_full_2021-2022.cs
```

Flower Check sheets
– Organize flowering data

Important Step!!
In the csv file, select all data with dates (visit dates, leaf sampling, and flower photos). Right click –> format cells. Select the date category –> YYYY-MM-DD. Save the csv and close it. This will standardize all dates to the same format and make working with them in R much easier!

```r
ACflr_df <- read_csv(file = here::here( "data_sheets", "season_end_flowerchecks_2022",  "AC_flowercheck_
```

```
## New names:
## * `Recipient` -> `Recipient...2`
## * `Plant_ID` -> `Plant_ID...3`
## * `F?` -> `F?...4`
## * `2022-04-09` -> `2022-04-09...5`
## * `2022-04-21` -> `2022-04-21...6`
## * `2022-05-03` -> `2022-05-03...7`
## * `2022-05-13` -> `2022-05-13...8`
## * `2022-05-25` -> `2022-05-25...9`
## * `2022-06-08` -> `2022-06-08...10`
## * `` -> `...11`
## * `` -> `...12`
## * `` -> `...13`
## * `LEAF` -> `LEAF...14`
## * `flr_P` -> `flr_P...15`
## * `Recipient` -> `Recipient...16`
## * `Plant_ID` -> `Plant_ID...17`
## * `F?` -> `F?...18`
## * `2022-04-09` -> `2022-04-09...19`
## * `2022-04-21` -> `2022-04-21...20`
## * `2022-05-03` -> `2022-05-03...21`
## * `2022-05-13` -> `2022-05-13...22`
## * `2022-05-25` -> `2022-05-25...23`
## * `2022-06-08` -> `2022-06-08...24`
## * `` -> `...26`
## * `` -> `...27`
## * `LEAF` -> `LEAF...28`
## * `flr_P` -> `flr_P...29`
```

```
## Warning: One or more parsing issues, call `problems()` on your data frame for details,
## e.g.:
##   dat <- vroom(...)
##   problems(dat)
```

```
## Rows: 1776 Columns: 29
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr   (6): Transect, Recipient...2, Recipient...16, F?...18, 2022-06-08...24...
## dbl   (2): Plant_ID...3, Plant_ID...17
```

```
## lgl   (6): F?...4, ...11, ...12, ...13, ...26, ...27
## date (15): 2022-04-09...5, 2022-04-21...6, 2022-05-03...7, 2022-05-13...8, 2...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
##Ignore 'F' plants - more up-to-date fitness plant data is matched in below

#Re-structure data: split groups
f1_AClong <- ACflr_df[c(1:15)] # Group 1: plants 1-5
f2_AClong <- ACflr_df[c(1, 16:29)] # Group 2: plants 6-10

#Revise names
colnames(f1_AClong) <- c("Transect", "Recipient", "Plant_ID", "F.", "Fl_4.9", "Fl_4.21", "Fl_5.3", "Fl_
colnames(f2_AClong) <- c("Transect", "Recipient", "Plant_ID", "F.", "Fl_4.9", "Fl_4.21", "Fl_5.3", "Fl_

#rbind, tidy up and remove unnecessary columns
f_AClong <- rbind(f1_AClong, f2_AClong) %>% arrange(Transect, Recipient, Plant_ID) %>% select(-c(F., D,


#Format recipient IDs for easier sorting - three digits (includes leading zeros)
f_AClong_2 <- f_AClong %>%
  mutate(Recipient_num = sprintf("%03d", (parse_number(Recipient)))) %>%
  mutate(Recipient = as.character(str_c("AC_", Recipient_num))) %>%
  select(Transect, Recipient, Recipient_num, Plant_ID, everything()) %>%
  select(-Recipient_num)


##Fix Dates
f_AClong_3 <- f_AClong_2 %>%
  mutate(Fl_4.21 = case_when(Fl_4.9 == "2022-04-22" ~ "2022-04-22",
                             Fl_4.21 == "2022-04-22" ~ "2022-04-22",
                             Fl_4.21 == "2022-04-21" ~ "2022-04-21")) %>%
  mutate(Fl_4.9 = case_when(Fl_4.9 == "2022-04-09" ~ "2022-04-09",
                            TRUE ~ as.character(NA))) %>%
  mutate(Fl_6.22 = case_when(Fl_6.22 == 19165 ~ "2022-06-22",
                             TRUE ~ as.character(NA)))
```

Combine Germination and Flowering data frames

```
#Merge by Recipient, Transect, and Plant ID
AC_GermFlow_1 <- left_join(AC_germ_full, f_AClong_3, by = c("Recipient", "Transect", "Plant_ID")) %>%
  mutate(Fl_6.22 = as_date(Fl_6.22),
         Fl_4.9 = as_date(Fl_4.9),
         Fl_4.21 = as_date(Fl_4.21)) #correct this column to be class 'date'
#Double-check that observation (row) numbers match up
```

Fitness Plant data

- Indicate which plants were fitness plants, including those replicated in the watering experiment

```
AC_Fplants <- read_csv(here::here("data_sheets", "fitness_plants_7-21-2022", "AC_F_plants.csv"), col_typ
  rename(F_plant = `SINGLE Fitness Plant`, F_multi = `ALL Fitness Plants`, F_plant_notes = NOTES) %>%
  select(Transect:F_plant_notes)

#Adjust Recipient names to match three-digit format
```

```
AC_Fplants2 <- AC_Fplants %>%
  mutate(Recipient_num = sprintf("%03d", (parse_number(Recipient)))) %>%
  mutate(Recipient = as.character(str_c("AC_", Recipient_num))) %>%
  select(-Recipient_num) %>%
  mutate(Replicated = case_when(Replicated == "R" ~ "R",
                                is.na(Replicated) ~ "n")) #change NAs to 'n' (no) for non-rep'd Recipie
```

Index fitness plants & Replicated (y/n) in germination and flowering data

```
# Merge the two dataframes - F plant data will be repeated x10
# Create new row (Plant ID) specific variables for main and replicated fitness plants
AC_GermFlow_2 <- merge(AC_GermFlow_1, AC_Fplants2) %>%
  mutate(FitP = case_when(F_plant == Plant_ID ~ TRUE,
                          F_plant != Plant_ID ~ FALSE)) %>%
  mutate(Rep_FitP = case_when(Replicated == "R" & Plant_ID != 1 & Plant_ID == as.numeric(str_match(F_mul
                              Replicated == "R" & Plant_ID == 1 & Plant_ID == as.numeric(str_match(F_mul
                              TRUE ~ FALSE)) %>%  #if else, set value to FALSE
  mutate(any_FitP = case_when(FitP == TRUE | Rep_FitP == TRUE ~ TRUE, #plant is any form of fitness pla
                              TRUE ~ FALSE))
#Note - added as.numeric() around the str_match calls above since an R update caused the previous versi

#Tidy up
AC_GermFlow_3 <- AC_GermFlow_2 %>%
  select(Block, Transect, Sequence, Donor, Recipient, Replicated, Plant_ID, FitP, Rep_FitP, any_FitP, e
  arrange(Transect, Sequence, Recipient, Plant_ID)
```

Fruit and Seed counts

Due to the structure of excel data sheets, it seems easiest to upload fruit and seed count data separately since different spreadsheets were used to record each. Info on plants' Block, Transect, sequence, etc. will be pulled from the fruit count sheets. As typos were discovered in fruit and seed counts, corrections were made in batches.

Updated: **Feb 05, 2023**

```
# Data on *when* fruits were collected will be omitted here; only the total fruit counts (closed, open,

AC_fruitprod_1 <- read_csv(here::here("data_sheets", "fruit_seed_counts_final", "AC-fruit-production_20
  select(1:11) #omit sow and flower date, seed count, and date-specific fruit counts by column index

AC_fruitprod_2 <- AC_fruitprod_1 %>%
  select(-c(Recipient, 9)) %>%
  rename(Recipient = `Recipient for Sorting`, Plant_ID = `Plant number of fitness plant (when chosen)`,

#Note: total fruits do not include eaten/damaged fruit



# Seed count data (viable seeds)
# only seed count data are retained from this spreadsheet to minimize errors
AC_seedct_1 <- read_csv(here::here("data_sheets", "fruit_seed_counts_final", "AC-seed-count_2023-02-05.
  select(-c(3, 6, 7, 9:29)) %>%
  rename(Recipient = `Recipient for Sorting`, Plant_ID = `Plant number of fitness plant (when chosen)`,


#merge data
```

```r
AC_fruitseed_ct1 <- left_join(AC_fruitprod_2, AC_seedct_1, by = c("Sequence", "Transect", "Recipient",


# summary variables, tidying up
AC_fruitseed_ct2 <- AC_fruitseed_ct1 %>%
  select(Transect, Sequence, Recipient, Plant_ID, closed_fruits_F, open_fruits_F, damaged_fruits_F, tota
  arrange(Transect, Sequence, Recipient)

## NOTE: total number of non-fitness fruits per segment may be under-represented here since not all cou
```

Seed Weights

Note: Prior to csv reading: - header row for seed weights was moved down to line up with other headers

```r
# Read in only # seeds weighed and total weight
# Calculate mean weight per seed

AC_seedmass_1 <- read_csv(here::here("data_sheets", "seed_mass", "AC_Seed_Mass_2023-02-05.csv"), skip =
```

```
## New names:
## Rows: 546 Columns: 25
## -- Column specification
## -------------------------------------------------------- Delimiter: "," chr
## (4): Transect, Recipient, Recipient for Sorting, ...14 dbl (21): Sequence,
## Plant number of fitness plant (when chosen), ...6, ...7,...
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## * `` -> `...6`
## * `` -> `...7`
## * `` -> `...8`
## * `` -> `...9`
## * `` -> `...10`
## * `` -> `...11`
## * `` -> `...12`
## * `` -> `...13`
## * `` -> `...14`
## * `` -> `...15`
## * `` -> `...16`
## * `` -> `...17`
## * `` -> `...18`
## * `` -> `...19`
## * `` -> `...20`
## * `` -> `...21`
## * `` -> `...22`
## * `` -> `...23`
## * `` -> `...24`
## * `` -> `...25`
```

```r
colnames(AC_seedmass_1) <- c("Sequence", "Transect", "Recipient_old", "Recipient", "Plant_ID", "closed_


AC_seedmass_2 <- AC_seedmass_1 %>%
  select(-starts_with(c("closed_fruits_", "viable_seeds_")), -Recipient_old) %>%
  mutate(msm_d1 = seed_mass_mg_d1/seeds_weighed_d1,
         msm_d2 = seed_mass_mg_d2/seeds_weighed_d2,
```

```
         msm_d3 = seed_mass_mg_d3/seeds_weighed_d3,
         msm_d4 = seed_mass_mg_d4/seeds_weighed_d4,
         msm_d5 = seed_mass_mg_d5/seeds_weighed_d5)


AC_seedmass_3 <- AC_seedmass_2 %>%
  mutate(msm_all = rowMeans(select(AC_seedmass_2, starts_with("msm_")), na.rm = TRUE)) %>%
  rowwise() %>%
  mutate(sd_msm_all = sd(c_across(c(msm_d1:msm_d5)), na.rm = TRUE))


##Merge with fruit/seed count data frame

AC_fruitseed_ct3 <- merge(AC_fruitseed_ct2, AC_seedmass_3, by = c("Transect", "Sequence", "Recipient",
#all.x means that any fitness plant x ID combos that don't match up won't be included.
```

Merge everything together:

```
AC_merged_df1 <- merge(AC_GermFlow_3, AC_fruitseed_ct3, by = c("Transect", "Sequence", "Recipient", "Pla
  select(Block, Transect, Sequence, Donor, Recipient, Replicated, Plant_ID, FitP, Rep_FitP, any_FitP, ev
  arrange(Block, Transect, Sequence, Recipient, Plant_ID) %>%
  mutate(closed_fruits_F = as.numeric(closed_fruits_F),
         total_fruits_F = as.numeric(total_fruits_F)) #correct counts to numeric values


#Correct Fruit NAs (but not seed count NAs) to 0s in cases where no fruit data was entered (fitness and
AC_merged_df2 <- AC_merged_df1 %>%
  mutate(closed_fruits_F = replace_na(closed_fruits_F, 0),
         total_fruits_F = replace_na(total_fruits_F, 0),
         seed_ct = replace_na(seed_ct, 0)) #REPLACE ALL seed_ct NAs with 0


#Calculate mean seeds per fruit, total_est_seed prod
#variables renamed later
AC_merged_df3 <- AC_merged_df2 %>%
  mutate(mean_seeds_per_fruit = seed_ct/closed_fruits_F,
         total_est_seed_production = mean_seeds_per_fruit*total_fruits_F)
#More fitness variables calculated later


#mean seeds per fruit and total est fruit production have NAs/NaNs for when there were no closed fruits,
```

Skeleton Weights & Corolla area and diameter

Load and merge-in these data sheets

```
##
skel_weights_AC <- read_csv(here::here("data_sheets", "skeleton_and_stem_data", "skeleton_weights_AC_202
  filter(!grepl("EXCLUDE", Notes)) %>% ## Filter out notes with "Exclude"
  rename(Plant_ID = `Fitness Plant_ID`, AG_biomass_mg = `Skeleton Weights (mg)`)
```

```
## New names:
## Rows: 441 Columns: 8
## -- Column specification
## -------------------------------------------------------- Delimiter: "," chr
## (5): Recipient, Units, Notes, Chosen for Pulverization? (y/n), ...8 dbl (3):
```

```
## Transect, Fitness Plant_ID, Skeleton Weights (mg)
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## * `` -> `...8`
```

```
#Watered points
skel_weights_AC_W <- read_csv(here::here("data_sheets", "skeleton_and_stem_data", "skeletons_watered_AC_
  select(-`Units (mg)`) %>%
  rename(Plant_ID = `Plant ID`, AG_biomass_mg = `Skeleton Weights`)
```

```
## Rows: 123 Columns: 6
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (3): Transect, Recipient, Notes
## dbl (2): Plant ID, Skeleton Weights
## lgl (1): Units (mg)
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
#Combine
skel_weights_all <- rbind(skel_weights_AC, skel_weights_AC_W) %>%
  select(-Notes) %>%
  filter(!is.na(AG_biomass_mg))


#Merge with full data set
AC_merged_df4 <- merge(AC_merged_df3, skel_weights_all, by = c("Transect", "Recipient", "Plant_ID"), all

#includes NF plants... how many?
AC_NFskels <- AC_merged_df4 %>%
  filter(!is.na(AG_biomass_mg)) %>%
  arrange(FitP)
#65/433 are Replicated Fitness plants, not main F plants
```

Corolla measurements

```
corollas_AC <- read_csv(here::here("data_sheets", "corolla_measures", "corollameasures_AC_2023-02-28.csv
  rename(photo_date = 4, corolla_diam_mm = 5, corolla_area_mm2 = 6, notes_corolla = 7) %>%
  filter(!grepl("REMOVE", notes_corolla)) ## Filter out notes with "REMOVE"
```

```
## New names:
## Rows: 3552 Columns: 25
## -- Column specification
## -------------------------------------------------------- Delimiter: "," chr
## (12): Transect, Recipient, URL of JPEG of Flower, Units...8, Units...10... dbl
## (10): Sequence, Plant_ID, Corolla Diameter, Corolla Area, (paste here),... lgl
## (2): ...24, Area of smallest polygon that encloses leaf tips and roset... date
## (1): Date_Flower_Photo_Taken_yr-mo-day
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## * `Units` -> `Units...8`
## * `Units` -> `Units...10`
## * `` -> `...15`
## * `` -> `...16`
## * `` -> `...17`
```

```
## * `` -> `...18`
## * `` -> `...19`
## * `Units` -> `Units...22`
## * `` -> `...24`
```

```
#Ignore warning; offending columns not selected

corollas_AC2 <- corollas_AC %>%
  filter(!is.na(corolla_diam_mm)) %>%
  mutate(photo_date = as_date(photo_date)) %>% #correct missing dates
  mutate(Recipient_num = sprintf("%03d", (parse_number(Recipient)))) %>% #Fix Recipient names
  mutate(Recipient = as.character(str_c("AC_", Recipient_num))) %>%
  select(-Recipient_num) %>%
  arrange(Transect, Recipient, Plant_ID, photo_date) %>%
  distinct(Transect, Recipient, Plant_ID, .keep_all = TRUE) #Keep only the earliest photo date data for
#Only about 6 duplicate observations (same plant, different dates) removed


## Merge with main data
AC_merged_df5 <- merge(AC_merged_df4, corollas_AC2, by = c("Transect", "Recipient", "Plant_ID"), all.x =


#check: includes NF plants...
AC_NFcorollas <- AC_merged_df5 %>%
  filter(!is.na(corolla_diam_mm)) %>%
  arrange(FitP)
#245/582 are Replicated Fitness & NF plants, not main F plants (158 totally NF)
```

Leaf Area & Weight

```
leaves_AC <- read_csv(here::here("data_sheets", "leaf_data", "AC_SLA_data_20230522.csv")) %>%
  rename(leaf_area_mm2 = 5, leafarea_notes = 6, leaf_dryweight_mg = 7, leafweight_notes = 8) %>%
  filter(!grepl("EXCLUDE", leafarea_notes)) %>% ## Filter out notes with "EXCLUDE"
  filter(!is.na(leaf_area_mm2)) %>% #exlcude NA leaf areas?
  mutate(leaf_area_mm2 = as.numeric(leaf_area_mm2))
```

```
## Rows: 399 Columns: 8
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (7): Block, Transect, Recipient, Plant_ID, Leaf area (mm^2), Area notes,...
## dbl (1): Dry weight (mg)
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
## Warning: There was 1 warning in `mutate()`.
## i In argument: `leaf_area_mm2 = as.numeric(leaf_area_mm2)`.
## Caused by warning:
## ! NAs introduced by coercion
```

```
#NA coercion for "No photo" note entered in area column

#Eliminate Duplicates
#Randomly select one leaf to be analyzed per transect x maternal ID combo
leaves_AC_duped <- leaves_AC %>%
  mutate(duped1 = duplicated(select(leaves_AC, c(Block, Transect, Recipient))),
```

```r
         duped2 = duplicated(select(leaves_AC, c(Block, Transect, Recipient)), fromLast = TRUE),
         duped = case_when(duped1|duped2 == TRUE ~ TRUE,
                           TRUE ~ FALSE)) %>% #highlight all duplicates
  select(-c(duped1, duped2))
# Select dupe rows that have both area and weight!


#This seems to work the best!
leaves_AC_unique <- leaves_AC_duped %>%
  group_by(Block, Transect, Recipient) %>%
  slice_sample(n = 1) %>%  #select a random row for each  combo of block, transect, recipient
  select(-c(duped, Plant_ID)) %>%  #Don't need plant ID...these were all non-fitness plants
  mutate(SLA_SEG = leaf_area_mm2 / leaf_dryweight_mg,
         LMA_SEG = 1 / SLA_SEG) %>% #Following line added: SEG indicates a segment-wide estimate
  rename(leaf_area_mm2_SEG = leaf_area_mm2,
         leaf_dryweight_mg_SEG = leaf_dryweight_mg)


#Merge in to main data
AC_merged_df6 <- merge(AC_merged_df5, leaves_AC_unique, by = c("Block", "Transect", "Recipient"), all.x


#
##Include plant-level leaf measurements as well:
leaves_AC_ind <- leaves_AC %>%
  mutate(Plant_ID = case_when(Plant_ID == "No ID" ~ NA,
                              TRUE ~ Plant_ID)) %>%  #Fix missing ID names
  filter(!is.na(Plant_ID)) %>% #remove obs missing ID (2)
  select(Block, Transect, Recipient, Plant_ID, leaf_area_mm2, leaf_dryweight_mg) %>% #ignore notes
  mutate(SLA = leaf_area_mm2 / leaf_dryweight_mg,
         LMA = 1 / SLA)

#Merge - sheet 6b
AC_merged_df6b <- merge(AC_merged_df6, leaves_AC_ind, by = c("Block", "Transect", "Recipient", "Plant_I

# make sure all measurements merged in
AC_NFleaves <- AC_merged_df6b %>%
  filter(!is.na(leaf_dryweight_mg)) %>%
  arrange(FitP)
# Should be all/mostly NF
```

Error checks:

```r
##Check for mismatched (fitness) plants

AC_Echeck1 <- AC_merged_df6b %>%
  filter(any_FitP == FALSE & total_fruits_F != 0) #do the same for seed count, using `>` and `!=`

AC_Echeck2 <- AC_merged_df6b %>%
  filter(any_FitP == TRUE) %>%
  filter(closed_fruits_F == 0 & seed_ct > 0)

##NA/NaN/Info values for mean seeds/est seed prod
AC_Echeck3 <- AC_merged_df6b %>%
```

```
  filter(any_FitP == TRUE & is.na(as.numeric(mean_seeds_per_fruit)))

AC_Echeck4 <- AC_merged_df6b %>%
  filter(any_FitP == TRUE & !is.finite(mean_seeds_per_fruit))


## Check for NA seed counts for fitness plants (when fruit production > 0)
#Will be empty when seed_ct NAs are replaced with 0
AC_Echeck5 <- AC_merged_df6b %>%
  filter(any_FitP == TRUE & is.na(seed_ct))
##NAs present here before setting seed_ct NAs to 0 were from fitness plants that had 0 closed fruit

#Seed mass
AC_Echeck5 <- AC_merged_df6b %>%
  filter(any_FitP == FALSE & !is.na(msm_all))
```

Trait calculations

```
AC_merged_df_fl <- AC_merged_df6b %>%
  mutate(FFD = do.call(pmin, c(select(., starts_with('Fl_')), na.rm = TRUE)), #First flowering date
         days_germ2flower = as.numeric(FFD - Germ_Date), #days from germination to flowering
         days_sow2flower = as.numeric(FFD - Sow_Date), #days from lab sowing to flowering
         days_plant2flower = as.numeric(FFD - Plant_Date), #days from field planting to flowering
         LFD = do.call(pmax, c(select(., starts_with('Fl_')), na.rm = TRUE)), #last flowering date
         fl_duration = as.numeric(LFD - FFD)) #Flowering duration
```

Phenology Revise Calculations of Fitness Data to match Year 2

```
#Updating Fruit count columns
#All Fitness plants (any kind) should have non-NA values for fruit and seed counts
AC_merged_df_flB <- AC_merged_df_fl %>%
  mutate(closed_fruits_F = case_when(any_FitP == TRUE & is.na(closed_fruits_F) ~ 0,
                                     any_FitP == TRUE & !is.na(closed_fruits_F) ~ closed_fruits_F,
                                     TRUE ~ closed_fruits_F),
         open_fruits_F = case_when(any_FitP == TRUE & is.na(open_fruits_F) ~ 0,
                                   any_FitP == TRUE & !is.na(open_fruits_F) ~ open_fruits_F,
                                   TRUE ~ open_fruits_F),
         total_fruits_F = case_when(any_FitP == TRUE & is.na(total_fruits_F) ~ 0,
                                   any_FitP == TRUE & !is.na(total_fruits_F) ~ total_fruits_F,
                                   TRUE ~ total_fruits_F))


#Survival measures
AC_merged_df_flC <- AC_merged_df_flB %>%
  mutate(surv_to_germ = case_when(!is.na(Germ_Date) ~ 1, #plants that germinated
                                  is.na(Germ_Date) ~ 0),
         surv_to_flower = case_when(!is.na(FFD) ~ 1, #plants that survived to flower
                                  is.na(FFD) ~ 0),
         surv_to_fruitprod = case_when(any_FitP == TRUE & total_fruits_F > 0 ~ 1, #plants that survived
                                       any_FitP == TRUE & total_fruits_F == 0 ~ 0),
         prod_closed_fruits = case_when(any_FitP == TRUE & closed_fruits_F > 0 ~ 1,
                                        any_FitP == TRUE & closed_fruits_F == 0 ~ 0),
         prod_filled_seeds = case_when(any_FitP == TRUE & seed_ct > 0 ~ 1, #plants that survived to pro
```

```
                                        any_FitP == TRUE & seed_ct == 0 ~ 0))  #else = 0


#Calculate Fecundity here
AC_merged_df_fl2 <- AC_merged_df_flC %>%
  rename(est_fecundity = total_est_seed_production) %>% #rename to match Y2
  mutate(across(c(est_fecundity, mean_seeds_per_fruit), ~ as.numeric(na_if(., NaN)))) %>% #Converts to
  mutate(mean_seeds_per_fruit = case_when(any_FitP == TRUE & closed_fruits_F == 0 ~ NA, #NA when 0 clos
                                          TRUE ~ mean_seeds_per_fruit)) %>%  #else, original value
  mutate(est_fecundity = case_when(any_FitP == TRUE & closed_fruits_F == 0 & open_fruits_F == 0 ~ 0,
                                   any_FitP == TRUE & closed_fruits_F == 0 & open_fruits_F
                                   any_FitP == TRUE & !is.na(est_fecundity) ~  est_fecundit
```

d13C data

```
Nemo_d13C <- read_csv(here::here("data_sheets", "isotope_data", "d13C_2022.csv")) %>%
  select(1,2,3,5,6,9) %>%
  rename(sample_ID = 1, d13C = 2, totalC_ug = 3, sample_weight_mg = 6)
#Save as UTF-8 file from excel to avoid import error

#filter to AC samples
Nemo_d13C_AC <- Nemo_d13C %>%
  filter(grepl('AC', sample_ID))

#unravel sample ID
Nemo_d13_AC2 <- Nemo_d13C_AC %>%
  mutate(Recipient = gsub('-', '_', substr(sample_ID, 1, 6)),
         Transect = as.numeric(str_remove(substr(sample_ID, 8, 10), "^0+")),
         Plant_ID = as.numeric(substr(sample_ID, 12,13))) %>%
  select(Transect, Recipient, Plant_ID, d13C, sample_weight_mg) #un-un-select Plant_ID


#Merge
AC_merged_df_fl3 <- merge(AC_merged_df_fl2, Nemo_d13_AC2, by = c("Recipient", "Transect", "Plant_ID"),


#check that all data made it (266 obs)
#testdf <- AC_merged_df_fl3 %>%
#  filter(!is.na(d13C))

#Check d13C for F/NF plants...
AC_NFd13C <- AC_merged_df_fl3 %>%
  filter(!is.na(d13C)) %>%
  arrange(FitP)
#includes 10 Rep Fs; rest main F


#Create segment-wide estimate of d13C:
Nemo_d13_AC2b <- Nemo_d13_AC2 %>%
  rename(d13C_SEG = d13C) %>% #rename for seg-wide measure
  select(Transect, Recipient, d13C_SEG) #un-select Plant_ID, dryweight of sample
```

```
AC_merged_df_fl3b <- merge(AC_merged_df_fl3, Nemo_d13_AC2b, by = c("Recipient", "Transect"), all.x = TRU
#Merge by Recipient and Transect only: segment-wide estimates of d13C since some plants were NF
```

Segment-wide estimates for skeleton weight, corolla area & diameter Skeleton Weight

```
#Merge to identify MAIN fitness plants
skel_weights_seg1 <- merge(AC_GermFlow_3, skel_weights_all, by = c("Recipient", "Transect", "Plant_ID")
  filter(!is.na(AG_biomass_mg)) %>%
  select(Block, Transect, Sequence, Donor, Recipient, Replicated, Plant_ID, FitP, Rep_FitP, any_FitP, AG


#Highlight ALL DUPES (multiple measures within a segment)
skel_weights_duped <- skel_weights_seg1 %>%
  mutate(duped1 = duplicated(subset(skel_weights_seg1, select = c("Block", "Transect", "Sequence", "Dono
         duped2 = duplicated(subset(skel_weights_seg1, select = c("Block", "Transect", "Sequence", "Dono
         duped = case_when(duped1|duped2 == TRUE ~ TRUE,
                           TRUE ~ FALSE))


#Filter out Replicated Plants when a Main plant is available (non-watered transects only)
#Note: Non-main FitPs (replicated) retained when only when a single measure taken from a segment, but d
skel_weights_unduped1 <- skel_weights_duped %>%
  filter(case_when(duped == TRUE & Replicated == "R" & Block != "W" ~ FitP == TRUE,
                   TRUE ~ any_FitP == TRUE)) %>% #keep all main FitPs when duped+replicated,else keep a
  filter(Block != "W")


#Randomly selected F plants - one measure per available segment
skel_weights_seg_RAND <- skel_weights_duped %>%
  group_by(Block, Transect, Recipient) %>%
  slice_sample(n = 1) %>%  #select a random row for each  combo of block, transect, recipient
  select(-c(duped, Plant_ID))  #%>%
 # select(Block:Recipient) #temp filter
#Confirms # obs = # segments (310)


#Anti-join to see missing obs
skel_weights_unduped_comp <- anti_join(skel_weights_seg_RAND, skel_weights_unduped1, by = c("Block", "Tr
  select(-c(Replicated:any_FitP, duped1, duped2))

#Main plant measures - remove unnecessary columns
skel_weights_unduped1b <- skel_weights_unduped1 %>%
  select(-c(Replicated:any_FitP, duped, duped1, duped2)) #Make sure columns match up

#Combine with missing complement - measurements from replicated/NF plants where no main plant available
skel_weights_SEG <- rbind(skel_weights_unduped1b, skel_weights_unduped_comp) %>%
  rename(AG_biomass_mg_SEG = AG_biomass_mg)
#End result: One measure per segment (from which a skeleton was collected), with main fitness plants re


#Merge with main data sheet
AC_merged_df_fl4 <- merge(AC_merged_df_fl3b, skel_weights_SEG, by = c("Block", "Transect", "Sequence",
```

Corolla Diameter & Area

Use same approach as above

```
#Merge to identify MAIN fitness plants
corollas_seg1 <- AC_merged_df5 %>%
  filter(!is.na(corolla_diam_mm)) %>% #diameter prioritized - main trait to measure (& fewer NAs)
  select(Block, Transect, Sequence, Donor, Recipient, Replicated, Plant_ID, FitP, Rep_FitP, any_FitP, c


#Highlight duplicates
corollas_duped <- corollas_seg1 %>%
  mutate(duped1 = duplicated(subset(corollas_seg1, select = c("Block", "Transect", "Sequence", "Donor",
         duped2 = duplicated(subset(corollas_seg1, select = c("Block", "Transect", "Sequence", "Donor",
         duped = case_when(duped1|duped2 == TRUE ~ TRUE,
                           TRUE ~ FALSE))


#Filter out Replicated Plants when a Main plant is available (non-watered transects only)
#Note: Duplicates (2+) of replicated plants are removed; merged back in after random selection below
corollas_unduped1 <- corollas_duped %>%
  filter(case_when(duped == TRUE & Replicated == "R" & Block != "W" ~ FitP == TRUE,
                   TRUE ~ any_FitP == TRUE)) %>% #keep all main FitPs when duped+replicated, else keep
  filter(Block != "W") #241 obs


#Randomly selected F plants - one measure per available segment
corollas_seg_RAND <- corollas_duped %>%
  group_by(Block, Transect, Recipient) %>%
  slice_sample(n = 1) %>%  #select a random row for each  combo of block, transect, recipient
  select(-c(duped, Plant_ID)) #%>%
  #filter(Block != "W")
##309 total obs; 267 obs excluding W block - number of segment measurements there should be


#Anti-join to get missing obs
corollas_unduped_comp <- anti_join(corollas_seg_RAND, corollas_unduped1, by = c("Block", "Transect", "S
  select(-c(Replicated:any_FitP, duped1, duped2))


#Main plant measures - remove unnecessary columns
corollas_unduped1b <- corollas_unduped1 %>%
  select(-c(Replicated:any_FitP, duped, duped1, duped2)) #Make sure columns match up

#Combine with missing complement - measurements from replicated/NF plants where no main plant available
corollas_SEG <- rbind(corollas_unduped1b, corollas_unduped_comp) %>%
  rename(corolla_diam_mm_SEG = corolla_diam_mm,
         corolla_area_mm2_SEG = corolla_area_mm2) #rename vars
#double-check that obs total as expected


#Merge
AC_merged_df_fl5 <- merge(AC_merged_df_fl4, corollas_SEG, by = c("Block", "Transect", "Sequence", "Dono
```

Duplicate & Final Error Check

```r
#All rows
AC_merged_df_fl5 %>%
  select(Recipient:Plant_ID) %>%
  get_dupes()
```

```
## No variable names specified - using all columns.
```

```
##      Recipient Plant_ID dupe_count
## 1       AC_034        1          5
## 2       AC_034        1          5
## 3       AC_034        1          5
## 4       AC_034        1          5
## 5       AC_034        1          5
## 6       AC_034        2          5
## 7       AC_034        2          5
## 8       AC_034        2          5
## 9       AC_034        2          5
## 10      AC_034        2          5
## 11      AC_034        3          5
## 12      AC_034        3          5
## 13      AC_034        3          5
## 14      AC_034        3          5
## 15      AC_034        3          5
## 16      AC_034        4          5
## 17      AC_034        4          5
## 18      AC_034        4          5
## 19      AC_034        4          5
## 20      AC_034        4          5
## 21      AC_034        5          5
## 22      AC_034        5          5
## 23      AC_034        5          5
## 24      AC_034        5          5
## 25      AC_034        5          5
## 26      AC_034        6          5
## 27      AC_034        6          5
## 28      AC_034        6          5
## 29      AC_034        6          5
## 30      AC_034        6          5
## 31      AC_034        7          5
## 32      AC_034        7          5
## 33      AC_034        7          5
## 34      AC_034        7          5
## 35      AC_034        7          5
## 36      AC_034        8          5
## 37      AC_034        8          5
## 38      AC_034        8          5
## 39      AC_034        8          5
## 40      AC_034        8          5
## 41      AC_040        1          5
## 42      AC_040        1          5
## 43      AC_040        1          5
## 44      AC_040        1          5
## 45      AC_040        1          5
## 46      AC_040        2          5
```

```
## 47       AC_040        2           5
## 48       AC_040        2           5
## 49       AC_040        2           5
## 50       AC_040        2           5
## 51       AC_040        3           5
## 52       AC_040        3           5
## 53       AC_040        3           5
## 54       AC_040        3           5
## 55       AC_040        3           5
## 56       AC_040        4           5
## 57       AC_040        4           5
## 58       AC_040        4           5
## 59       AC_040        4           5
## 60       AC_040        4           5
## 61       AC_040        5           5
## 62       AC_040        5           5
## 63       AC_040        5           5
## 64       AC_040        5           5
## 65       AC_040        5           5
## 66       AC_040        6           5
## 67       AC_040        6           5
## 68       AC_040        6           5
## 69       AC_040        6           5
## 70       AC_040        6           5
## 71       AC_040        7           5
## 72       AC_040        7           5
## 73       AC_040        7           5
## 74       AC_040        7           5
## 75       AC_040        7           5
## 76       AC_040        8           5
## 77       AC_040        8           5
## 78       AC_040        8           5
## 79       AC_040        8           5
## 80       AC_040        8           5
## 81       AC_055        1           5
## 82       AC_055        1           5
## 83       AC_055        1           5
## 84       AC_055        1           5
## 85       AC_055        1           5
## 86       AC_055        2           5
## 87       AC_055        2           5
## 88       AC_055        2           5
## 89       AC_055        2           5
## 90       AC_055        2           5
## 91       AC_055        3           5
## 92       AC_055        3           5
## 93       AC_055        3           5
## 94       AC_055        3           5
## 95       AC_055        3           5
## 96       AC_055        4           5
## 97       AC_055        4           5
## 98       AC_055        4           5
## 99       AC_055        4           5
## 100      AC_055        4           5
```

```
## 101      AC_055         5           5
## 102      AC_055         5           5
## 103      AC_055         5           5
## 104      AC_055         5           5
## 105      AC_055         5           5
## 106      AC_055         6           5
## 107      AC_055         6           5
## 108      AC_055         6           5
## 109      AC_055         6           5
## 110      AC_055         6           5
## 111      AC_055         7           5
## 112      AC_055         7           5
## 113      AC_055         7           5
## 114      AC_055         7           5
## 115      AC_055         7           5
## 116      AC_055         8           5
## 117      AC_055         8           5
## 118      AC_055         8           5
## 119      AC_055         8           5
## 120      AC_055         8           5
## 121      AC_058         1           5
## 122      AC_058         1           5
## 123      AC_058         1           5
## 124      AC_058         1           5
## 125      AC_058         1           5
## 126      AC_058         2           5
## 127      AC_058         2           5
## 128      AC_058         2           5
## 129      AC_058         2           5
## 130      AC_058         2           5
## 131      AC_058         3           5
## 132      AC_058         3           5
## 133      AC_058         3           5
## 134      AC_058         3           5
## 135      AC_058         3           5
## 136      AC_058         4           5
## 137      AC_058         4           5
## 138      AC_058         4           5
## 139      AC_058         4           5
## 140      AC_058         4           5
## 141      AC_058         5           5
## 142      AC_058         5           5
## 143      AC_058         5           5
## 144      AC_058         5           5
## 145      AC_058         5           5
## 146      AC_058         6           5
## 147      AC_058         6           5
## 148      AC_058         6           5
## 149      AC_058         6           5
## 150      AC_058         6           5
## 151      AC_058         7           5
## 152      AC_058         7           5
## 153      AC_058         7           5
## 154      AC_058         7           5
```

```
## 155      AC_058         7           5
## 156      AC_058         8           5
## 157      AC_058         8           5
## 158      AC_058         8           5
## 159      AC_058         8           5
## 160      AC_058         8           5
## 161      AC_070         1           5
## 162      AC_070         1           5
## 163      AC_070         1           5
## 164      AC_070         1           5
## 165      AC_070         1           5
## 166      AC_070         2           5
## 167      AC_070         2           5
## 168      AC_070         2           5
## 169      AC_070         2           5
## 170      AC_070         2           5
## 171      AC_070         3           5
## 172      AC_070         3           5
## 173      AC_070         3           5
## 174      AC_070         3           5
## 175      AC_070         3           5
## 176      AC_070         4           5
## 177      AC_070         4           5
## 178      AC_070         4           5
## 179      AC_070         4           5
## 180      AC_070         4           5
## 181      AC_070         5           5
## 182      AC_070         5           5
## 183      AC_070         5           5
## 184      AC_070         5           5
## 185      AC_070         5           5
## 186      AC_070         6           5
## 187      AC_070         6           5
## 188      AC_070         6           5
## 189      AC_070         6           5
## 190      AC_070         6           5
## 191      AC_070         7           5
## 192      AC_070         7           5
## 193      AC_070         7           5
## 194      AC_070         7           5
## 195      AC_070         7           5
## 196      AC_070         8           5
## 197      AC_070         8           5
## 198      AC_070         8           5
## 199      AC_070         8           5
## 200      AC_070         8           5
## 201      AC_081         1           5
## 202      AC_081         1           5
## 203      AC_081         1           5
## 204      AC_081         1           5
## 205      AC_081         1           5
## 206      AC_081         2           5
## 207      AC_081         2           5
## 208      AC_081         2           5
```

```
## 209      AC_081        2           5
## 210      AC_081        2           5
## 211      AC_081        3           5
## 212      AC_081        3           5
## 213      AC_081        3           5
## 214      AC_081        3           5
## 215      AC_081        3           5
## 216      AC_081        4           5
## 217      AC_081        4           5
## 218      AC_081        4           5
## 219      AC_081        4           5
## 220      AC_081        4           5
## 221      AC_081        5           5
## 222      AC_081        5           5
## 223      AC_081        5           5
## 224      AC_081        5           5
## 225      AC_081        5           5
## 226      AC_081        6           5
## 227      AC_081        6           5
## 228      AC_081        6           5
## 229      AC_081        6           5
## 230      AC_081        6           5
## 231      AC_081        7           5
## 232      AC_081        7           5
## 233      AC_081        7           5
## 234      AC_081        7           5
## 235      AC_081        7           5
## 236      AC_081        8           5
## 237      AC_081        8           5
## 238      AC_081        8           5
## 239      AC_081        8           5
## 240      AC_081        8           5
## 241      AC_082        1           5
## 242      AC_082        1           5
## 243      AC_082        1           5
## 244      AC_082        1           5
## 245      AC_082        1           5
## 246      AC_082        2           5
## 247      AC_082        2           5
## 248      AC_082        2           5
## 249      AC_082        2           5
## 250      AC_082        2           5
## 251      AC_082        3           5
## 252      AC_082        3           5
## 253      AC_082        3           5
## 254      AC_082        3           5
## 255      AC_082        3           5
## 256      AC_082        4           5
## 257      AC_082        4           5
## 258      AC_082        4           5
## 259      AC_082        4           5
## 260      AC_082        4           5
## 261      AC_082        5           5
## 262      AC_082        5           5
```

```
## 263     AC_082        5          5
## 264     AC_082        5          5
## 265     AC_082        5          5
## 266     AC_082        6          5
## 267     AC_082        6          5
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## 3419    AC_211        4          3
## 3420    AC_211        4          3
## 3421    AC_211        5          3
## 3422    AC_211        5          3
## 3423    AC_211        5          3
## 3424    AC_211        6          3
## 3425    AC_211        6          3
## 3426    AC_211        6          3
## 3427    AC_211        7          3
## 3428    AC_211        7          3
## 3429    AC_211        7          3
## 3430    AC_211        8          3
## 3431    AC_211        8          3
## 3432    AC_211        8          3
## 3433    AC_211        9          3
## 3434    AC_211        9          3
## 3435    AC_211        9          3
## 3436    AC_211       10          3
## 3437    AC_211       10          3
## 3438    AC_211       10          3
## 3439    AC_212        1          3
## 3440    AC_212        1          3
## 3441    AC_212        1          3
## 3442    AC_212        2          3
## 3443    AC_212        2          3
## 3444    AC_212        2          3
## 3445    AC_212        3          3
## 3446    AC_212        3          3
## 3447    AC_212        3          3
## 3448    AC_212        4          3
```

```
## 3449    AC_212        4          3
## 3450    AC_212        4          3
## 3451    AC_212        5          3
## 3452    AC_212        5          3
## 3453    AC_212        5          3
## 3454    AC_212        6          3
## 3455    AC_212        6          3
## 3456    AC_212        6          3
## 3457    AC_212        7          3
## 3458    AC_212        7          3
## 3459    AC_212        7          3
## 3460    AC_212        8          3
## 3461    AC_212        8          3
## 3462    AC_212        8          3
## 3463    AC_212        9          3
## 3464    AC_212        9          3
## 3465    AC_212        9          3
## 3466    AC_212       10          3
## 3467    AC_212       10          3
## 3468    AC_212       10          3
## 3469    AC_213        1          3
## 3470    AC_213        1          3
## 3471    AC_213        1          3
## 3472    AC_213        2          3
## 3473    AC_213        2          3
## 3474    AC_213        2          3
## 3475    AC_213        3          3
## 3476    AC_213        3          3
## 3477    AC_213        3          3
## 3478    AC_213        4          3
## 3479    AC_213        4          3
## 3480    AC_213        4          3
## 3481    AC_213        5          3
## 3482    AC_213        5          3
## 3483    AC_213        5          3
## 3484    AC_213        6          3
## 3485    AC_213        6          3
## 3486    AC_213        6          3
## 3487    AC_213        7          3
## 3488    AC_213        7          3
## 3489    AC_213        7          3
## 3490    AC_213        8          3
## 3491    AC_213        8          3
## 3492    AC_213        8          3
## 3493    AC_213        9          3
## 3494    AC_213        9          3
## 3495    AC_213        9          3
## 3496    AC_213       10          3
## 3497    AC_213       10          3
## 3498    AC_213       10          3
## 3499    AC_215        1          3
## 3500    AC_215        1          3
## 3501    AC_215        1          3
## 3502    AC_215        2          3
```

```
## 3503    AC_215        2          3
## 3504    AC_215        2          3
## 3505    AC_215        3          3
## 3506    AC_215        3          3
## 3507    AC_215        3          3
## 3508    AC_215        4          3
## 3509    AC_215        4          3
## 3510    AC_215        4          3
## 3511    AC_215        5          3
## 3512    AC_215        5          3
## 3513    AC_215        5          3
## 3514    AC_215        6          3
## 3515    AC_215        6          3
## 3516    AC_215        6          3
## 3517    AC_215        7          3
## 3518    AC_215        7          3
## 3519    AC_215        7          3
## 3520    AC_215        8          3
## 3521    AC_215        8          3
## 3522    AC_215        8          3
## 3523    AC_215        9          3
## 3524    AC_215        9          3
## 3525    AC_215        9          3
## 3526    AC_215       10          3
## 3527    AC_215       10          3
## 3528    AC_215       10          3
## 3529    AC_216        9          3
## 3530    AC_216        9          3
## 3531    AC_216        9          3
## 3532    AC_216       10          3
## 3533    AC_216       10          3
## 3534    AC_216       10          3
## 3535    AC_217        9          3
## 3536    AC_217        9          3
## 3537    AC_217        9          3
## 3538    AC_217       10          3
## 3539    AC_217       10          3
## 3540    AC_217       10          3
## 3541    AC_218        9          3
## 3542    AC_218        9          3
## 3543    AC_218        9          3
## 3544    AC_218       10          3
## 3545    AC_218       10          3
## 3546    AC_218       10          3
```

```
#Fitness Plants
AC_merged_df_fl5 %>%
  filter(any_FitP == TRUE) %>%
  get_dupes()
```

```
## No variable names specified - using all columns.
```

```
## No duplicate combinations found of: Block, Transect, Sequence, Donor, Recipient, Plant_ID, Replicate
```

```
##  [1] Block                 Transect              Sequence
##  [4] Donor                 Recipient             Plant_ID
```

```
##  [7] Replicated            FitP                  Rep_FitP
## [10] any_FitP              Germ_Date             Sow_Date
## [13] Plant_Date            Fl_4.9                Fl_4.21
## [16] Fl_5.3                Fl_5.13               Fl_5.25
## [19] Fl_6.8                Fl_6.22               LEAF
## [22] flr_P                 F_multi               F_plant
## [25] F_plant_notes         closed_fruits_F       open_fruits_F
## [28] damaged_fruits_F      total_fruits_F        nonfitness_fruits
## [31] seed_ct               seeds_weighed_d1      seed_mass_mg_d1
## [34] seeds_weighed_d2      seed_mass_mg_d2       seeds_weighed_d3
## [37] seed_mass_mg_d3       seeds_weighed_d4      seed_mass_mg_d4
## [40] seeds_weighed_d5      seed_mass_mg_d5       msm_d1
## [43] msm_d2                msm_d3                msm_d4
## [46] msm_d5                msm_all               sd_msm_all
## [49] mean_seeds_per_fruit  est_fecundity         AG_biomass_mg
## [52] photo_date            corolla_diam_mm       corolla_area_mm2
## [55] notes_corolla         leaf_area_mm2_SEG     leafarea_notes
## [58] leaf_dryweight_mg_SEG leafweight_notes      SLA_SEG
## [61] LMA_SEG               leaf_area_mm2         leaf_dryweight_mg
## [64] SLA                   LMA                   FFD
## [67] days_germ2flower      days_sow2flower       days_plant2flower
## [70] LFD                   fl_duration           surv_to_germ
## [73] surv_to_flower        surv_to_fruitprod     prod_closed_fruits
## [76] prod_filled_seeds     d13C                  sample_weight_mg
## [79] d13C_SEG              AG_biomass_mg_SEG     corolla_diam_mm_SEG
## [82] corolla_area_mm2_SEG  dupe_count
## <0 rows> (or 0-length row.names)
```

```r
#Main fitness plants
AC_merged_df_fl5 %>%
  filter(FitP == TRUE) %>%
  get_dupes()
```

```
## No variable names specified - using all columns.
##
## No duplicate combinations found of: Block, Transect, Sequence, Donor, Recipient, Plant_ID, Replicate
```

```
##  [1] Block                 Transect              Sequence
##  [4] Donor                 Recipient             Plant_ID
##  [7] Replicated            FitP                  Rep_FitP
## [10] any_FitP              Germ_Date             Sow_Date
## [13] Plant_Date            Fl_4.9                Fl_4.21
## [16] Fl_5.3                Fl_5.13               Fl_5.25
## [19] Fl_6.8                Fl_6.22               LEAF
## [22] flr_P                 F_multi               F_plant
## [25] F_plant_notes         closed_fruits_F       open_fruits_F
## [28] damaged_fruits_F      total_fruits_F        nonfitness_fruits
## [31] seed_ct               seeds_weighed_d1      seed_mass_mg_d1
## [34] seeds_weighed_d2      seed_mass_mg_d2       seeds_weighed_d3
## [37] seed_mass_mg_d3       seeds_weighed_d4      seed_mass_mg_d4
## [40] seeds_weighed_d5      seed_mass_mg_d5       msm_d1
## [43] msm_d2                msm_d3                msm_d4
## [46] msm_d5                msm_all               sd_msm_all
## [49] mean_seeds_per_fruit  est_fecundity         AG_biomass_mg
## [52] photo_date            corolla_diam_mm       corolla_area_mm2
```

```
## [55] notes_corolla         leaf_area_mm2_SEG     leafarea_notes
## [58] leaf_dryweight_mg_SEG leafweight_notes      SLA_SEG
## [61] LMA_SEG               leaf_area_mm2         leaf_dryweight_mg
## [64] SLA                   LMA                   FFD
## [67] days_germ2flower      days_sow2flower       days_plant2flower
## [70] LFD                   fl_duration           surv_to_germ
## [73] surv_to_flower        surv_to_fruitprod     prod_closed_fruits
## [76] prod_filled_seeds     d13C                  sample_weight_mg
## [79] d13C_SEG              AG_biomass_mg_SEG     corolla_diam_mm_SEG
## [82] corolla_area_mm2_SEG  dupe_count
## <0 rows> (or 0-length row.names)
```

Master sheets (v1)

```
## First, create a Full (long) version with all data - one row per plant

AC_mastersheet_v1 <- AC_merged_df_fl5 %>%
  mutate(Year = 2022, Gen = "G1") %>% #label year and generation
  select(Year, Gen, Block, Transect, Sequence, Donor, Recipient, Replicated, Plant_ID, FitP, Rep_FitP, a
  arrange(Block, Transect, Sequence, Recipient, Plant_ID)

#Rename and organize variables to match Year 2 data:
AC_mastersheet_v2 <- AC_mastersheet_v1 %>%
  rename(F_plant_ID = F_plant, #number of F plant
         F_plant = FitP, #T/F is F plant
         filled_seeds = seed_ct,
         open_fruits = open_fruits_F,
         closed_fruits = closed_fruits_F,
         total_fruits = total_fruits_F,
         skel_dryweight_mg = AG_biomass_mg,
         skel_dryweight_mg_SEG = AG_biomass_mg_SEG) %>%
  mutate(est_fecundity_int = round(est_fecundity)) #round to nearest whole number


 #Add Greenhouse cohort
#2021 Tray position data sheets - extract and merge in greenhouse cohort data:
AC_tray21 <- read_csv(here::here("data_sheets", "tray_positions_2021_year1", "AC_tray_rand.csv"), skip =
  select(Cohort, Donor_ID, Recipient_ID) %>% #Block and Transect not necessary to save here
  mutate(Recipient_num = sprintf("%03d", (parse_number(Recipient_ID)))) %>% #Fix Recipient/Donor names
  mutate(Recipient = as.character(str_c("AC_", Recipient_num))) %>%
  mutate(Donor_num = sprintf("%03d", (parse_number(Donor_ID)))) %>%
  mutate(Donor = as.character(str_c("AC_", Donor_num))) %>%
  select(-c(Donor_num, Recipient_num, Donor_ID, Recipient_ID)) %>%
  distinct() #remove duplicate cases


#merge
AC_mastersheet_v3 <- left_join(AC_mastersheet_v2, AC_tray21, by = c("Donor", "Recipient")) %>%
  select(-c(F_plant_notes, Fl_4.9:flr_P, damaged_fruits_F, seeds_weighed_d1:msm_d5, photo_date, notes_c
  select(Year, Sequence, Cohort, Block, Transect, Donor, Recipient, Gen, Replicated, Plant_ID, F_plant:
```

Write CSVs

```r
#Save full (long) version
write_csv(AC_mastersheet_v3, here::here("data_sheets", "compiled_sheets", "AC_mastersheet_full_2022.csv
# has records of all plants (including non-fitness)


#
## Fitness Plants Only

AC_mastersheet_F_mains <-  AC_mastersheet_v3 %>%
  filter(F_plant == TRUE) #Only main fitness plants

AC_mastersheet_F_all <- AC_mastersheet_v3 %>%
  filter(any_FitP == TRUE) #main and multiple fitness plants (IDs replicated in watering experiment)


write_csv(AC_mastersheet_F_mains, here::here("data_sheets", "compiled_sheets", "AC_mastersheet_Fitness-

write_csv(AC_mastersheet_F_all, here::here("data_sheets", "compiled_sheets", "AC_mastersheet_Fitness-al
```